Proposal for

Computer Vision PROJECT
Umm Al Qura University

Project ID: 3

Project Title: Smart Parking detection with proximity recommendations

Team Members:

Student Name	Student ID	Email	Role
Asail Hamdan Al-Osaimi	445000051	s445000051@uqu.edu.sa	Leader
Reema Salem Alharbi	445000219	s445000219@uqu.edu.sa	Member
Enas Abdullah Saud	44512190	s44512190@uqu.edu.sa	Member
Noran Abdullah Aljodi	445005178	s445005178@uqu.edu.sa	Member
Rema Saed Althaqfi	445005754	s445005754@uqu.edu.sa	Member

Start Date: 1/2/2026

End Date: 7/6/2026

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• Implemented modules/components

At this stage, the dataset preparation and preprocessing module has been successfully implemented using Python, OpenCV, and Google Colab. The PKLot dataset was downloaded and explored to understand its structure, including training, validation, and testing folders.

A subset of approximately 6000 images was sampled from the training dataset to reduce computational complexity and improve processing efficiency during development. The sampled images were organized into a separate dataset for preprocessing and experimentation.

The preprocessing pipeline was implemented to prepare the images for subsequent computer vision tasks. This module includes:

-Reading and loading parking images from the dataset

-Filtering unreadable or corrupted images

-Resizing all images to a fixed resolution of 128×128 pixels

-Organizing and exporting the processed images into a clean dataset structure

In addition, the cleaned dataset was saved into a compressed dataset file and prepared for future stages such as HOG feature extraction and occupancy classification using machine learning algorithms.

• Preliminary results

The preprocessing stage successfully generated a clean and standardized parking image dataset suitable for feature extraction and classification tasks. Approximately 6000 parking images were sampled and processed from the original PKLot dataset.

The implemented preprocessing pipeline successfully resized all valid images to a consistent format while filtering unreadable images. Visual inspection of the processed dataset confirmed that image quality was preserved after preprocessing and that the images remained suitable for parking occupancy analysis.

The cleaned dataset is now fully prepared for the next stages of the project, including:

-HOG-based feature extraction

-Parking occupancy classification using SVM and KNN

-Further evaluation and performance analysis

Initial experiments confirmed that the preprocessing workflow operates correctly and efficiently within the Google Colab environment.

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- ## Code Explanation Step by Step with Screenshots of Outputs

Technical Challenges and Solutions

1. Total Failure of Traditional Detection Methods (HOG)

- **Challenge**: Initial attempts to implement the **HOG (Histogram of Oriented Gradients)** algorithm yielded no functional results. The algorithm failed to produce any detections or meaningful output when applied to the parking dataset, proving it completely incapable of handling the complexity of the task.
- **Solution**: Recognizing this total failure, the project immediately pivoted to a **Deep Learning** approach using **YOLOv8**. This shift was essential as it moved from a failing manual feature extraction method to a robust, automated neural network capable of processing the data successfully.

2. Hardware Constraints and Computational Demands

- **Challenge**: Training advanced computer vision models like YOLOv8 imposed a significant burden on the available hardware. The initial device lacked the necessary **GPU capabilities**, leading to excessive training times and extreme latency that hindered the development progress.
- **Solution**: To overcome this infrastructure bottleneck, the workload was migrated to a high-performance laptop equipped with an **NVIDIA GPU**. By leveraging **CUDA-accelerated** parallel processing, the training cycle was optimized.

3. Software Library Compatibility

- **Challenge**: Security updates in the software libraries introduced restrictive protocols that blocked the model from loading its pre-trained weights.
- **Solution**: This was resolved by implementing a **programmatic override**. A custom function wrapper was used to adjust the loading parameters, ensuring a seamless restoration of the model without disrupting the environment stability.

4. Optimization of Architectural Efficiency

- **Challenge**: Identifying a model configuration that delivers high precision while remaining computationally efficient for processing data.
- **Solution**: The **YOLOv8 Nano** variant was selected for its optimized architecture. It achieved a remarkable **mAP50 of 99.4%**, successfully balancing high precision with lightweight resource consumption.

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• Updated task distribution among team members

Student Name	Initial task
Enas Abdullah Saud	
Reema Salem Alharbi	
Rema Saed Althaqfi	
Noran Abdullah Aljodi	
Asail Hamdan Al-Osaimi	implementing the A* Search Algorithm in the system

References

[1] P. Almeida, L. S. Oliveira, E. Silva Jr., A. Britto Jr., and A. Koerich, "PKLot – A robust dataset for parking lot classification," Expert Systems with Applications, vol. 42, no. 11, pp. 4937–4949, 2015.

[2] A. M. N. Alhajali, "PKLot Dataset," Kaggle, 2023. [Online]. Available: https://www.kaggle.com/datasets/ammarnassanalhajali/pklot-dataset. [Accessed: Mar. 4, 2026].